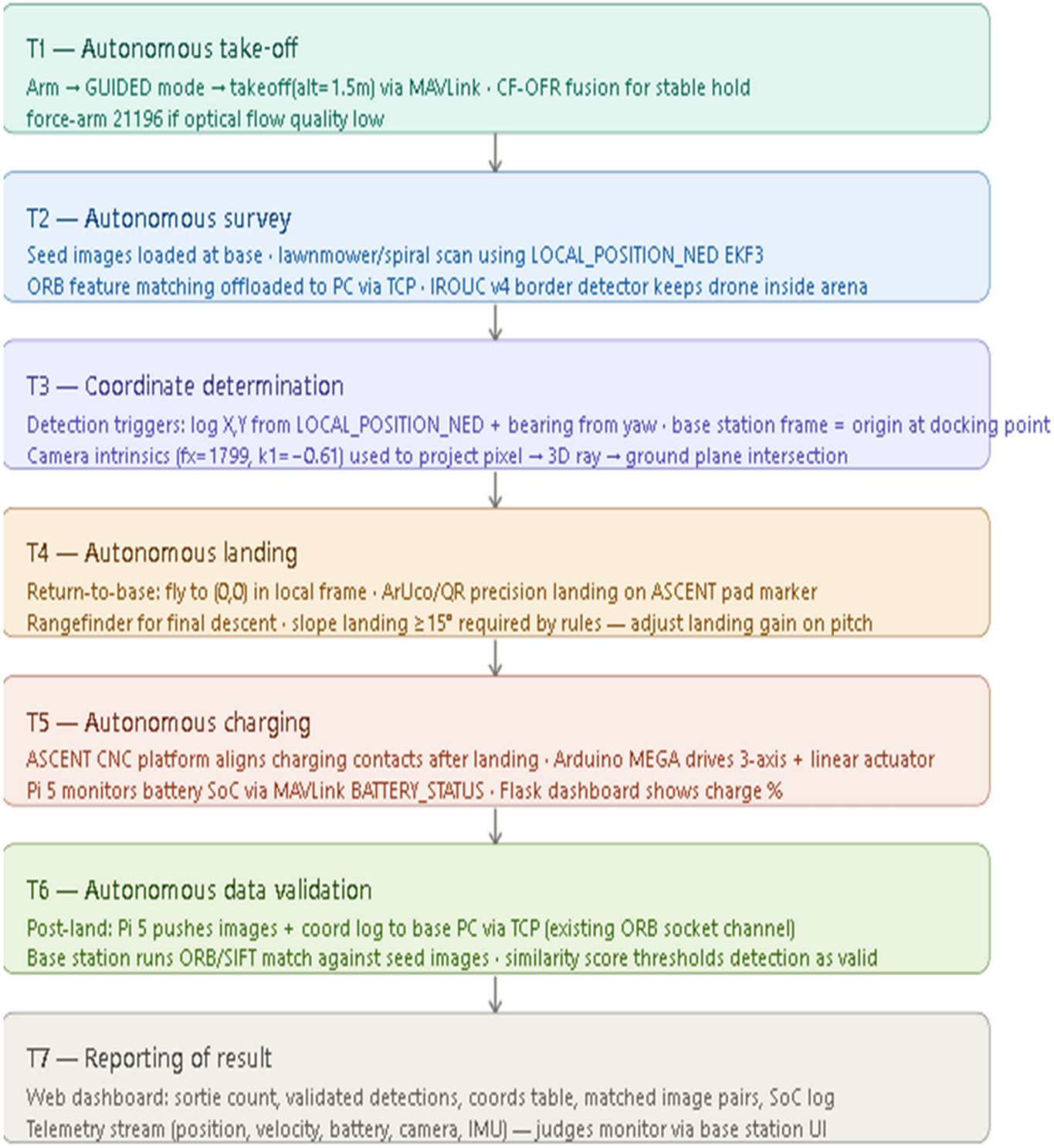
I.TASKS AND APPROACH:



T1-Autonomous Take-Off

T2-Autonomous Survey

T3-Coordinate Determination

T4-Autonomous Landing

T5-Autonomous Charging

T6-Autonomous Data Validation

T7-Reporting of Result

II. SERAPH ZENITH MARK 1-*Autonomous GPS-Denied Target Acquisition & Reconnaissance Drone*:

1. APPROACH:

- **Fully autonomous, closed-loop sortie**- arm, take-off, search, confirm, report, land - zero human-in-the-loop at runtime.
- **Coarse-to-fine search doctrine**-a viewpoint-invariant wide-FOV prior (LoFTR) answers *where to look*; a fast frame-level matcher (ORB) answer "*what is there"*. Detection compute is spent only where the prior says it matters.
- **Quorum-based confirmation**- no single frame is ever trusted. A target clears only on a >60% vote across independently scored candidates, making a one-off false lock structurally non-viable.
- **Decoupled landing phase**- precision touchdown runs on a dedicated fiducial marker (ArUco), independent of the search-time sensor suite, with yaw resolved before any lateral correction.
  Seraph Zenith (Qualifications) → Upgraded to Seraph Zenith Mark 1 (Eliminations).

2. KEY FEATURES:

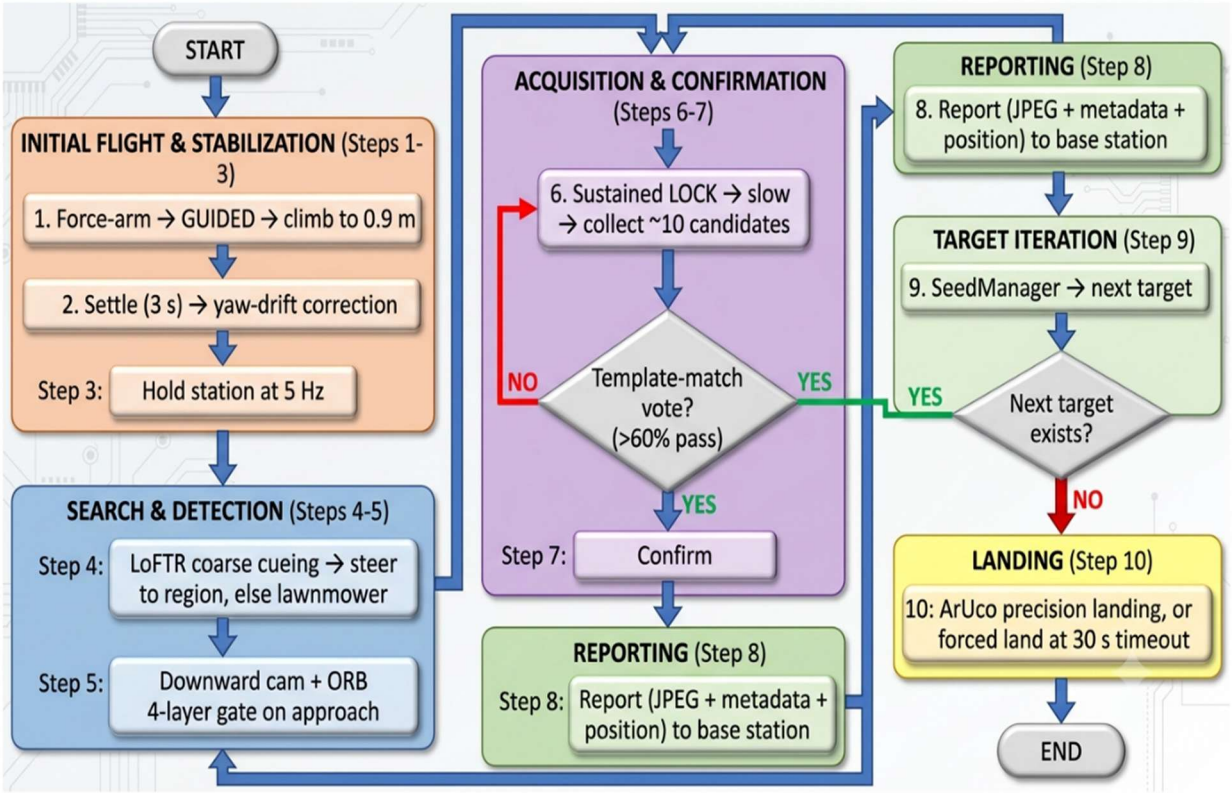
Capability	Seraph Zenith	Seraph Zenith Mark-1	Remarks
Take-off sequence	Generic actuation chain — FC → ESC (DShot600) → motors, 480 Hz loop.	Scripted force-arm (MAVLink bypass) → GUIDED → MAV_CMD_NAV_TAKEOFF to 0.9 m, with EKF settle + yaw-drift correction.**(software+ hardware)**	Concept → fully parameterized autonomous takeoff routine.
Search & detection	Dual-stage ORB (coarse) + LightGlue (fine) under "ATVS" — flight-state-gated matching still in development.	Dual-stage LoFTR (coarse cueing, wide ArduCam) + ORB (fine lock, downward cam) with a 4-layer security gate.**(software+ hardware)**	Proposal-stage gating logic now shipped as quantified, threshold-bound gate.
Target confirmation	Confidence logging proposed for post-mission audit; no formal voting threshold defined.	Quorum vote: >60% of ~10 collected candidates must pass 128×128 template match before confirmation.**(software)**	Logging concept matured into an auditable, statistically robust confirmation layer.
Landing	AutoSwap dock: downward camera + ultrasonic sensors detect pad (test scheduled).	Closed-loop ArUco (DICT_4X4_50) precision landing — yaw-first alignment, PID lateral servoing, proximity-scaled velocity, 30 s timeout → forced land.**(software+ hardware)**	Generic pad-detection concept → fully specified, failure-tolerant landing controller.
Reporting	Confidence/telemetry logging noted as a diagnostic goal.	Structured MSG_ACQUIRE transmission (JPEG + score + vote tally + timestamp), signal file, North/East position logging.**(software+ hardware)**	Diagnostic intent → operational, judge-ready acquisition record.
State estimation	GPS-denied VIO (OpenVINS → ArduPilot EKF) scheduled for validation.	Optical-flow (MTF-02P) velocity + rangefinder altitude actively fused into EKF during survey.**(software+ hardware)**	Flow-based fusion confirmed operational; VIO integration remains the next validation milestone.
Emergency response	Eight-point response matrix proposed (battery RTB, telemetry loss, RPi crash, jamming, etc.).	Landing-timeout failsafe (30 s → forced land) implemented and testable.**(software)**	One of eight proposed failsafe's now concretely instantiated; remainder pending round-wise validation.

Fig.1: Top and bottom mounts hardware configuration.



3. OPERATIONAL SEQUENCE:

Fig.2: Operational Sequence of Seraph Zenith Mark 1



III. ASCENT-*Autonomous Swap & Charging Exchange Technology*:

Drone Battery recharging Base Station

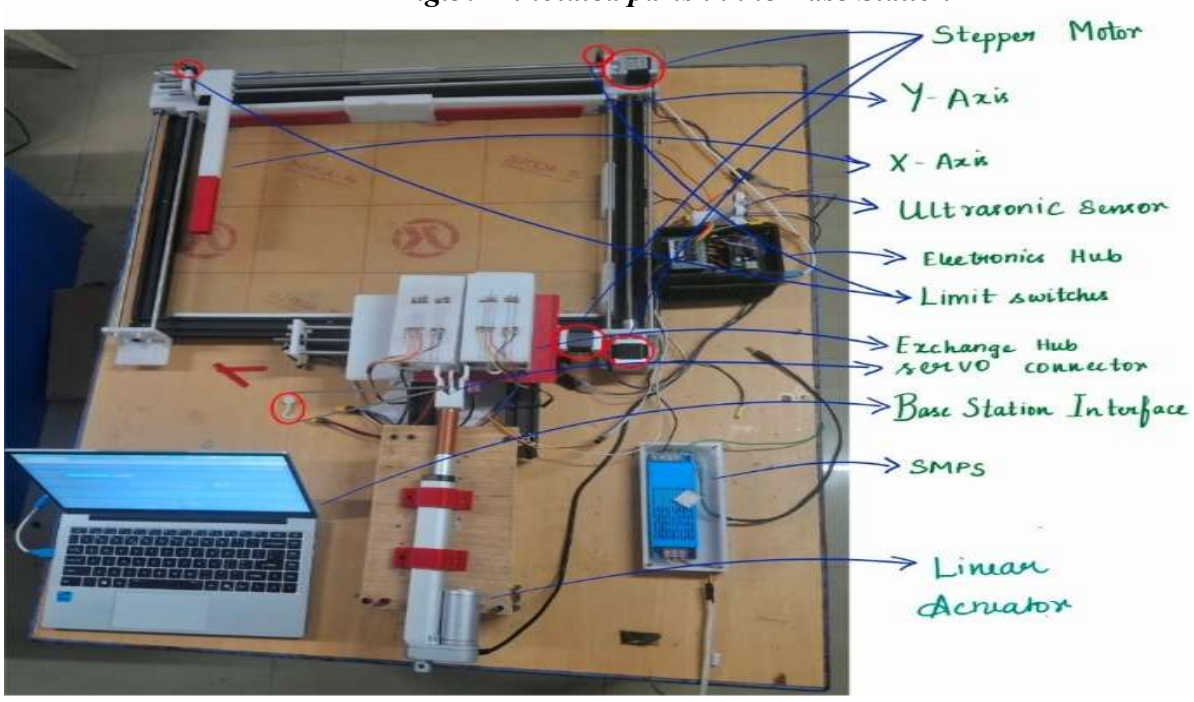
Ascent is an autonomous ground-based battery swap station designed for quadrotor drones operating in field survey and remote sensing missions. The system eliminates manual battery intervention by automatically detecting drone landing, performing battery swapping and prepare the drone for the next flight cycle - all without human involvement.

1. APPROACH:

ASCENT uses a 3-axis CNC-style positioning platform to align the battery swap mechanism with precision. An ultrasonic sensor detects drone arrival at the landing pad. Once confirmed, the X and Y axes retract to expose the swap zone, the linear actuator performs the battery exchange, and the servo operates the battery latch mechanism. After the swap is complete, the axes return to home position and the station enters a standby state ready for the next cycle.

2. KEY FEATURES:

Fig.3: Annotated parts in the Base Station



- Fully autonomous operation — no human intervention required
- Ultrasonic-based drone detection with 50cm threshold
- Minimum 7-minute inter-swap timer to prevent false triggers
- 3-axis stepper motor positioning (X, Y, Z axes)
- Linear actuator for battery push/pull mechanism
- Servo-controlled battery latch (0° locked, 90° released)
- 6 limit switches for precise axis homing and end-stop protection
- Real-time web dashboard via Flask over USB serial
- Infinite cycle capability with Z-axis alternating direction logic.

3. OPERATIONAL SEQUENCE:

Cycle 1 (First Swap):

- X and Y axes move forward to home positions (X HOME, Y FWD limit switches)
- Ultrasonic sensor scans for drone landing (threshold ≤50cm)
- 3-second confirmation wait after drone detected
- X axis reverses to X REV limit, Y axis reverses to Y REV limit
- Z axis moves forward to Z FWD limit switch
- Linear actuator extends for 25 seconds with servo at 0° (battery locked)
- Linear actuator retracts for 25 seconds with servo at 90° (battery released)
- Z axis checks pressed limit switch and moves to opposite limit
- Linear actuator extends and retracts again (25s each) without servo
- Y axis returns to Y FWD home, then X axis returns to X HOME
- Linear actuator performs final extend/retract cycle (25s each)
- Station enters standby — waits minimum 7 minutes before next swap

Cycle 2+ (Subsequent Swaps):

- From the **second cycle onwards**, the **X-axis and Y-axis remain at their home positions**.
- Each new cycle **starts directly with ultrasonic scanning**, eliminating the need for X and Y homing.
- The **Z-axis direction is determined by checking the currently active limit switch**.
- If one limit switch is pressed, the **Z-axis moves toward the opposite side**.
- This logic enables **bidirectional battery tray operation across consecutive cycles**.
- The system automatically alternates the battery tray movement direction, ensuring **continuous and efficient operation**.

Battery Recharging:



- Copper rails act as charging contacts drone battery slides onto rails on landing
- Rails connected directly to SkyRC B100 Neooutput terminals (positive and negative)
- SkyRC B100 Neo auto-detects Li-ion chemistry and begins charging automatically on contact
- No manual plug-in needed, rails make contact when battery is in position
- Balance lead connected separately to 100 Neo balance port for cell monitoring
- Charging begins immediately after battery is seated by the actuator